

OL7014 Integer Programming
Homework 2
Due Oct 6, start of class

Write down your solutions and submit them as a single PDF to loss0f0706@gmail.com.

Problem 1. Cadets and a married couple

(20 points)

Three cadets (A, B, C) are commissioned into three branches, one each, to maximize the total fit:

fit	Signal	Artillery	Engineers
cadet A	2	5	1
cadet B	5	3	2
cadet C	2	3	5

- (a) **(5 pts)** Formulate the IP and write it as $\max c^\top x$ s.t. $Ax = b$, $x \in \{0, 1\}^9$, with x flattened row by row. Write c , A , b with numbers.
- (b) **(4 pts)** Show that A is TU with the checklist (give M_1 , M_2). What does this say about the LP relaxation?
- (c) **(3 pts)** Solve the LP relaxation with `gurobipy` and confirm the solution is integral. Include a screenshot of your code and its output.
- (d) **(3 pts)** Cadets A and B are married, and policy forbids sending A to Artillery *and* B to Signal (both are overseas tours). Write this as one linear constraint.
- (e) **(5 pts)** Add the constraint and solve both the LP relaxation and the IP. Report both solutions and the gap, and explain where the fractions come from.

Problem 2. Absolute cap or relative cap?

(20 points)

Two drone teams fly x_1 (North) and x_2 (South) sorties, $x \in \mathbb{Z}_+^2$, to maximize $3x_1 + 2x_2$. The tanker carries fuel for at most 6 sorties in total. Two rules are proposed to keep North from taking all the fuel:

Version I $x_1 \leq 3$ (North flies at most 3 sorties)

Version II $x_1 - x_2 \leq 1$ (North flies at most one more than South)

Hand work only for this problem.

- (a) **(4 pts)** Write both IPs as $\max c^\top x$ s.t. $Ax \leq b$, $x \in \mathbb{Z}_+^2$ (tanker row first). Compute every square-submatrix determinant of A_I and A_{II} : which is TU? Why can a cap on a single variable, $x_j \leq u_j$, never destroy TU?
- (b) **(6 pts)** Draw both LP relaxations and mark their lattice points. Report z_{LP} and z_{IP} for each version and explain the gap using (a).
- (c) **(5 pts)** Keep A_{II} with an integer right-hand side $b = (b_1, b_2)$, $b_1 \geq b_2 \geq 0$. Solve for the corner where both rows are tight and show it is integral exactly when $b_1 + b_2$ is even. At $b = (7, 1)$ the LP optimum is integral — does that make A_{II} TU?
- (d) **(5 pts)** In Version I, North's real limit is 7 battery packs at 2 per sortie, and a staff officer writes $x_1 \leq 3.5$. Solve the LP relaxation. Which hypothesis of Hoffman–Kruskal fails? Fix the row by rounding.

Problem 3. Relief convoys

(20 points)

Relief trucks move from two sources through a hub to two divisions, whole truckloads only, at minimum cost. All roads are one-way.

node		b_i	road	(1, 3)	(1, 4)	(2, 3)	(2, 5)	(3, 4)	(3, 5)	(4, 5)
1	Port	+9	cost	3	7	2	6	2	3	2
2	Airfield	+5	capacity	8	4	5	3	7	6	3
3	Hub	0								
4	North	-8								
5	South	-6								

- (a) **(3 pts)** Which problem from Week 5 can model this problem? Draw the graph and define $G = (V, E)$.
- (b) **(4 pts)** Define your decision variable x . Write the corresponding matrix A and vectors c , b , and specify the dimension of each, e.g. $x \in \mathbb{R}^?$, $A \in \mathbb{R}^{? \times ?}$.
- (c) **(4 pts)** Explain why A is TU. Solve the LP relaxation with **gurobipy** and confirm that every flow is a whole truckload. Include a screenshot of your code and its output.
- (d) **(5 pts)** Roads (1, 3) and (3, 4) share a highway that carries at most 10 truckloads. Add this constraint, solve the LP relaxation and the IP, and report the gap. Find a 2×2 submatrix with determinant ± 2 that explains it.
- (e) **(4 pts)** Drop the highway rule. The Hub can instead receive at most 10 truckloads: model this by splitting the Hub into Hub-in and Hub-out, joined by a dock road of capacity 10. Is the new model still TU? Solve its LP relaxation. What is the modelling lesson compared with (d)?

Problem 4. The LAP

(20 points)

Two bases (1: North, 2: South) may hold a relief reserve for a town (3) with demand 8t and a village (4) with demand 4t. Opening costs are $f_1 = 20$, $f_2 = 16$. The five one-way roads cost, per ton, $c_{13} = 2$, $c_{23} = 4$, $c_{24} = 1$, $c_{34} = 1$, $c_{43} = 2$ (no capacities). Use the Week 6 LAP with $h_i = 1$, $u_i = 40$, $o_i = 1$, $B = 16$, $p = 1$, and linking rows $r_i \leq Bx_i$.

- (a) **(5 pts)** Write the LAP for this instance with numbers.
- (b) **(5 pts)** Which entries of the constraint matrix break TU? Show that once x is fixed, the rest of the model is TU. (Hint: read the budget row as the conservation row of one more node.)
- (c) **(6 pts)** Solve the IP and its LP relaxation with **gurobipy** and report both plans and the gap. Then fix x at the IP's value and solve the LP again: is the solution integral, and why?
- (d) **(4 pts)** Which variables are stage 1 and which are stage 2? Keep the IP's stage-1 decisions but let the town's demand be 10t. Solve stage 2: what is the total cost, and where does the shortage land?

Problem 5. Strong vs weak linking

(20 points)

Three candidate depots D1, D2, D3 and three brigades B1, B2, B3 sit on a ring road D1-B1-D2-B2-D3-B3-D1; each brigade can be supplied only by its two neighbouring depots. Let $x_i \in \{0, 1\}$ open depot D_i , and $y_{ij} \geq 0$ be the *fraction* of brigade B_j 's demand supplied by D_i (shipping cost $c_{ij}d_jy_{ij}$).

cost per ton	B1	B2	B3	opening cost
D1	1	—	2	20
D2	2	1	—	16
D3	—	2	1	18
demand (t)	4	5	6	

- (a) **(5 pts)** Write the *strong* formulation (one row $y_{ij} \leq x_i$ per road) and the *weak* formulation (one row $\sum_j y_{ij} \leq Mx_i$ per depot). What is the smallest valid M here?
- (b) **(4 pts)** Prove that the strong formulation is better, using the two-step recipe.
- (c) **(6 pts)** Solve the IP and both LP relaxations with **gurobipy**. Verify $z_{\text{LP}}^{\text{weak}} \leq z_{\text{LP}}^{\text{strong}} \leq z_{\text{IP}}$, and say which bound is more useful. Include a screenshot of your code and its output.
- (d) **(5 pts)** The strong LP is still fractional, so its matrix is not TU. Derive $x_i + x_{i'} \geq 1$ for each brigade's two depots; which 3×3 matrix from the Week 5 note do these rows form? Add them, divide by 2 and round up: does this cut close the gap?